

The Impacts of HTTPS Adoption for Online Services Located in China

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ABSTRACT

While HTTPS (one of the best-practiced internet security protocols) grows in popularity to the point of being nearly universally adopted by most modern web services, domains located in China appear to be lagging behind this world-wide trend. Security risks exist for websites which do not implement TLS under very strict conditions, and this can leave the hosts and users open to attacks while also introducing vulnerabilities for the unauthorized access of data.

A collection of tools was used within this study to automatically collect information on the protocols used in-practice for a large set of popular real-world websites. The collected information was analyzed to discern the security fidelity of each domain along a set of preselected categories. These data collected from domains in different geographic regions were compared to identify any quantitative differences between the internet security landscape inside and outside of China. This study found disparities in HTTPS adoption between China and other countries is still present in websites that are active today. However, substantial progress is being made to increase the security of the internet in China, so there is reason to believe the under-utilization of HTTPS will not last for much longer.

1 Introduction

Hypertext transfer protocol secure (or HTTPS) is a tool to provide a means for secure communication over the web. It combines the

Hypertext transfer protocol (HTTP) for data communication with Transport Layer Security (TLS) to achieve end-to-end encrypted data. While the adoption of this protocol has been widely praised for its innovation, domains specifically located within China have historically displayed lower-than-average rates of integrating HTTPS¹.

Prior work has been performed with the intention of accurately measuring different characteristics of the HTTPS online landscape and then attributing those characteristics to physical, geographical, political, cultural, and social phenomena and explanations². Additionally, the unique set of security features present for a particular online service may create vulnerabilities which also warrant analysis and discussion. The purpose of this study is to provide an updated, more recently collected dataset of online activity for analysis as well as expand on the discussion on the qualities of domains hosted within China. To achieve this, this study will occur within three primary phases.

Phase 1: Program applications to automatically request and acquire information related to HTTPS from popular domains across the web.

Phase 2: Perform analysis on measured information related to differences in HTTPS adoption between domains located in China vs. other countries.

¹ Comparisons include the HTTPS vs HTTP adoption statistics within the Cloudflare Radar online application [1].

² See the Related Work section for additional explanation.

Phase 3: Briefly explore the reasons for and impacts of HTTPS adoption on domains located in China.

2 Related Work

There has been considerable work to accurately quantify this phenomenon, and this study aims to build upon this existing collection. Some notable contributions in this field include the following two research papers.

Assessment of the impacts of TLS vulnerabilities in the HTTPS ecosystem in China describes a methodology to quantify the level of security present within a specific portion of the network [2]. Various potential attacks on domains not meeting certain TLS criteria were proposed to identify the biggest risks to internet security, especially within the region of China. The method of data acquisition in this study is influenced by some of the methods pursued within this 2018 paper. Additionally, with this background as context, changes made to the HTTPS ecosystem in China since 2018 were observed with the new data collected within this study.

Measuring HTTPS Adoption on the Web aims to compare the various security measurements in different networks to identify the differences between them [3]. While not explicitly mentioning China, this study takes a broader look at HTTPS adoption and explores regional differences as a main contributor for the influence of domains upgrading from HTTP to HTTPS. In addition, the study by Felt, et. al. explores different conditions for classifying websites supporting HTTPS including differentiating between extended page loads and strict page loads. This differentiation helped inspire the investigative methods used within this study.

Lastly, Cloudflare Radar, a powerful network reporting tool, also contains an excellent introduction to the type of statistical comparisons which provide the most value to this discussion. It measures the HTTPS utilization of servers in China at about 83% which is considerably lower than the 98% it reports worldwide. The measurements within this study were expected to see similar data.

3 Approach

The approach for the collection and analysis of network data for the purposes of calculating security fidelity involves three main components: the utilization of available tools to kickoff and automate the acquisition of data, the formation of specific requests to construct a viable set of sampleable sites, and the collection of available security-specific information to populate the dataset.

3.1 Overview of Tools

The successful acquisition of a complete dataset depended on the incorporation of many paramount software tools with access to the network.

In addition to being the initial inspiration for this study, Cloudflare Radar additionally contains useful statistics on web traffic which contributed significantly to the discussion on security metrics worth focusing on [1]. The Tranco dataset, which is a public dataset containing popular websites based on monitored web traffic, provided a good baseline from which to build the dataset used in this study [4]. Furthermore, the MaxMind GeoIP database delivers the capability to identify the geographic location for a server [5]. This is a critical feature for enabling the separation of domains located in China and those stationed elsewhere.

Along with these tools, the following Python modules were also leveraged:

- `requests`: enables the construction and delivery of HTTPS requests [6].
- `socket`: is an interface for the inter-process communication (particularly IPv4) [7].
- `ssl`: provides a TLS/SSL wrapper for socket objects [8].
- `OpenSSL`: is a cryptographic tool for analyzing TLS connections [9].
- `geoip2`: provides an API for querying the MaxMind GeoIP database [10].
- `multiprocessing`: enables process-based parallelism for faster execution [11].

3.2 Dataset Composition

A list of 4.6 million of the most popular domains in October and November 2023 was acquired from Tranco to provide an initial collection of sites comprising the majority of network activity. `socket` was then used to perform a DNS lookup and collect the IPv4 addresses from each of these domains starting at the top of the list provided by Tranco. This is an important step as the MaxMind GeoIP database requires an IP address (not just a URL domain name) in order to return a geographic location. GeoIP was then leveraged to automatically capture the country in which each of the domains was located. This way, websites contained within China can be compared against those located in other countries.

3.3 Security Measurements

From the baseline of domains being associated with IP addresses and geographic locations, further measurements for security features continued.

Each of the connections made, especially to websites outside of the US, took a handful of seconds on average. To counteract this, and receive the most data possible, the multiprocessing module was used to send and receive multiple requests simultaneously in parallel. The requests module in Python was used to identify which domains supported HTTPS. This was accomplished by first requesting an HTTPS version of each domain. If the result returned was HTTP and not HTTPS, it was determined that the site did not support compatibility with sending HTTPS responses, at least upon first connection.

`ssl`, `OpenSSL`, and `socket` were then used to acquire all of the TLS certificates from each of the compatible websites. The validity of each acquired certificate was confirmed to ensure up-to-date TLS compatibility, and the contents of the certificate were further used for analysis. Lastly, `ssl` and `socket` together identified the specific version of TLS that was being used during each of these connections (version 1.2 or 1.3).

4 Results and Analysis

The initial dataset, prior to any measurements over the web, was comprised of 4.6 million

domains. During each of the measurement steps there were instances in which some domains were found to be incompatible for later measurements. For example, if a site did not support HTTPS, then the step involving the acquisition of TLS certificates would not be able to be completed. Furthermore, other domains timed-out prior to a query being complete, and these domains were similarly filtered out from the dataset. To capture this, **Figure 1** below shows the number of domains which passed each of the steps.

Step	Number of Domains
Tranco import	4,644,227
IPv4 address recovery	271,614
Geolocation	229,717
HTTPS compatibility	111,318
TLS version check	35,144
TLS/SSL certificate	23,545

Figure 1: Number of domains which successfully passed each stage of data composition and security measuring.

Two steps in particular created a synthetic bottleneck for this study (the IPv4 acquisition step, and the TLS version check). Even with parallelism, each of these steps ran relatively slowly compared to their predecessors which reduced the overall sample size that fully completed each stage. Despite these challenges, a considerable sample size was still achieved from which reliable conclusions could be drawn.

Figure 2 below shows the distribution of domains for each of the most popular 11 countries. In this China can be seen to have a respectable sample-size of 6,400 domains which successfully passed the GeoIP step to determine server location. A total of 203 countries were represented within this dataset even though only the most popular 11 are displayed here.

Rank	Country	Initial Size
1	United States	119,120
2	Germany	13,382
3	Russia	12,982
4	The Netherlands	7,378
5	France	6,993
6	China	6,400
7	Canada	5,597
8	Japan	5,408
9	United Kingdom	5,229
10	Ireland	3,155
11	India	2,960

Figure 2: Number of domains which were successfully geolocated by country.

The next subsections will explore some of the characteristics and trends shared by domains in different countries.

4.1 HTTPS Utilization

The percentage of sites which utilized HTTPS instead of HTTP by country is presented within **Figure 3**.

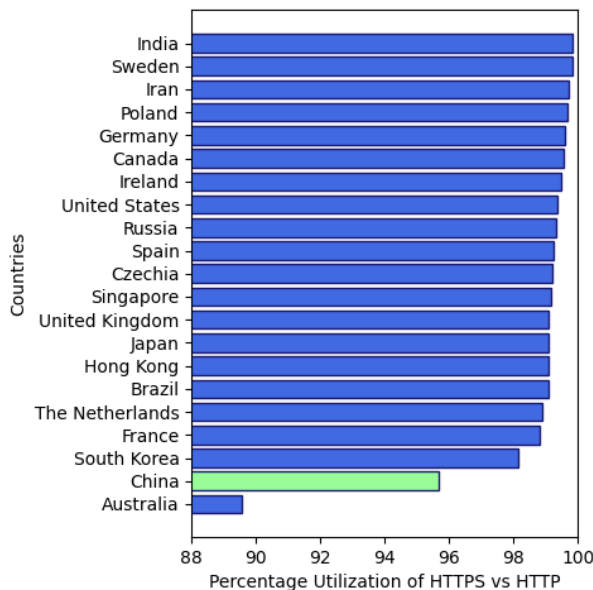


Figure 3: HTTPS Percentage Utilization by Country

As shown by this chart, domains located in China rank relatively low in HTTPS percentage utilization at 95.7%. Australia is found to be the only country in the collection plotted above which falls below the utilization percentage of China.

While the HTTPS utilization of other countries was performed, only countries represented by at least 500 unique domains are shown in **Figure 3**. This ensures that the statistics within this chart are relatively stable and are a reflection of a larger trend instead of being swayed by a few outliers.

4.2 TLS Version 1.3 Utilization

The percentage of sites which utilize TLS version 1.3 instead of TLS version 1.2 in HTTPS requests by country is presented within **Figure 4**.

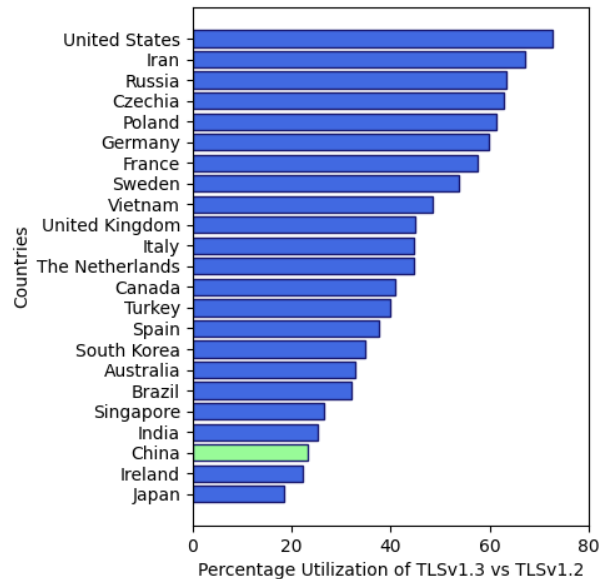


Figure 4: TLS version 1.3 Percentage Utilization by Country

It can be seen that domains located in China also rank relatively low in the adoption of TLS version 1.3 at around 23.2%. Ireland and Japan appear to be the only countries with a larger majority of domains using TLS version 1.2. While the calculations of TLS version 1.3 utilization percentage of other countries were performed, only countries represented by at least 100 unique domains are shown in **Figure 4**. As with the previous section, this is to ensure the percentages that were calculated are statistically reliable for comparing the sets of domains from each country.

4.3 Certificate Valid Duration

The average duration (that is the time between the starting date and the expiration date) of valid

TLS/SSL certificates by country is presented below in **Figure 5**.

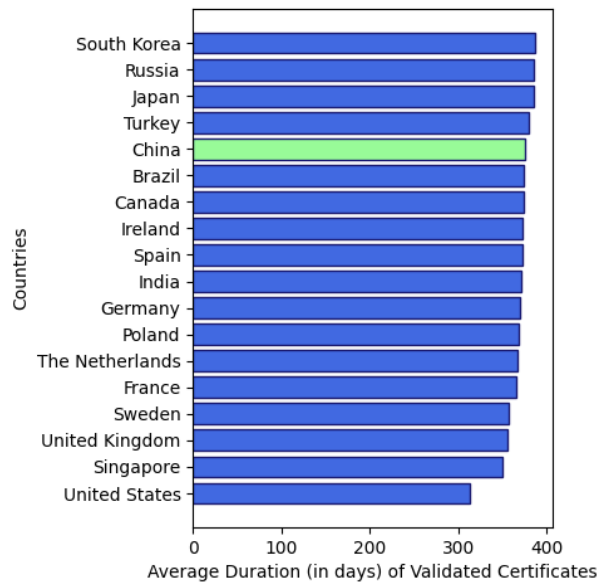


Figure 5: Average Valid Duration of TLS/SSL Certificates by Country

Contrary to the previous sections, within this section, China appears to be in the upper half of the websites within this chart insofar as it has a higher average duration of issued certificates. That being said, there are a few interesting features about this metric. First, there does not appear to be a significant spread in certificate duration. All domains, regardless of their country, appear to have around 1-year durations. This seems reasonable since if this is the typical duration, certificates would need to be updated about once every year.

It is also interesting to note that the United States is the only country with domains that average more than 1 month away (and below) the 1-year mean. While duration of TLS/SSL certificates does not directly result in more or less security (unless, for example, that longer certificates are vulnerable to obsolescence during their long lifetime) it is still an interesting finding within this study that might be better suited to be explored within a more focused study on certificate assignment.

4.4 Certificate Remaining Duration

The average time until expiration (that is the difference in time between when these certificates were acquired and when they expire) of valid TLS/SSL certificates by country is shown in **Figure 6**.

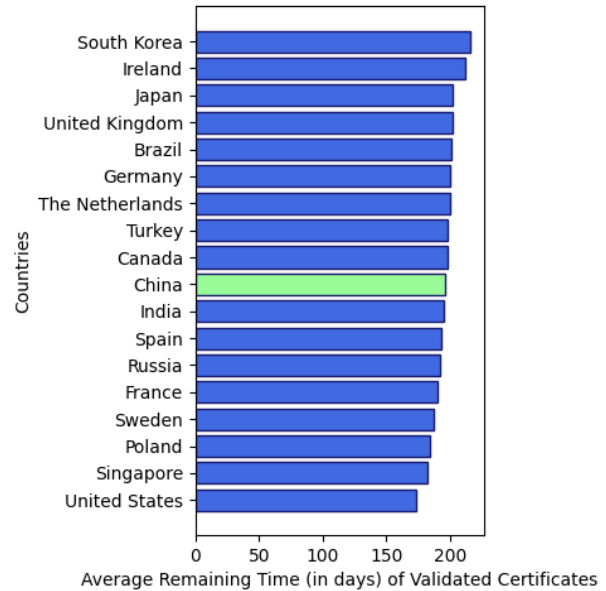


Figure 6: Average Time Until Expiration of TLS/SSL Certificates by Country

Domains in China sit right in the middle-of-the-pack in terms of the remaining time for their certificates to be valid. Similar to the total certificate duration shown in the previous section, the time until expiration for certificates does not appear to have any significant spread when examining the differences between countries. The average of all countries appears to be about 6 months. It seems that this data may be less revealing about the quality of the certificate issued while being more relevant for an investigation on the calendar timing in which most websites update their certificates.

4.5 Certificate Issuers

This section contains two charts to be compared with one another. **Figure 7** shows the number of sites within China that utilize certificate issuers from each country (US=United States, CN=China, etc.). Supplementary to this, **Figure 8** shows the number of sites worldwide which

leverage each of the major countries containing the certificate issuing entities.

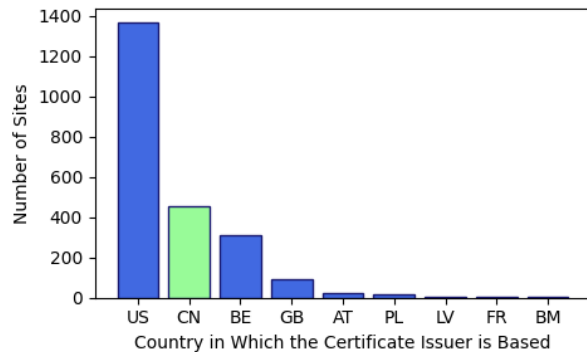


Figure 7: Websites in China: Number of Sites Utilizing Certificate Issuers Based in Each Country

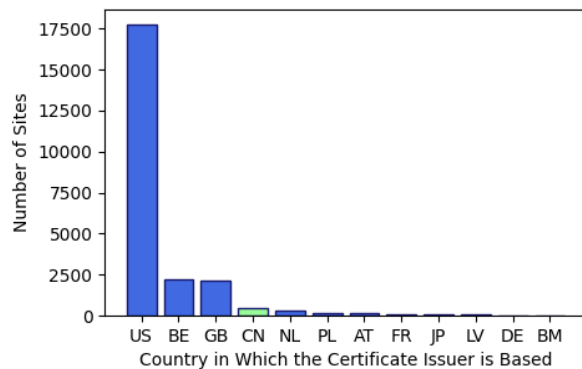


Figure 8: Websites Worldwide: Number of Sites Utilizing Certificate Issuers Based in Each Country

It is interesting to see that United States certificate issuers are the most utilized issuers, even for domains located in China. It is, however, still seen and expected that certificate issuers within China are the second most utilized within China. They are also more represented within China then in general across websites worldwide. It may be worth considering the challenges associated with partnering with a certificate issuer that is geographically very far from the webservice looking to implement HTTPS into their site. This may add to the low utilization in HTTPS seen for websites located within China.

5 Discussion

The measurements made within this study capture the previously documented phenomena of websites located within China containing a lower

utilization of HTTPS and a slower adoption to advancements in network security such as TLS version upgrades. However, comparing these measurements to some previous studies and resources reveal that the internet landscape within China is improving and is becoming more secure for hosts and users. This section will explore some of the reasons why the data shown in this study may be representative of real-world differences between countries, identify some of the security risks of low-HTTPS adoption, briefly examine a new wave of security enhancements to the internet in China, and propose some potential future work to expand upon this study.

5.1 Some Reasons for Disparity in HTTPS Adoption Statistics

First, there do appear to be many reasons why China in particular is seen to be an outlier in many of the measurements made during this study.

HTTPS adoption within China has had a distinct history as compared to other countries which has helped drive the results in this study. As explored within “Timeline of HTTPS Adoption,” there exists historical reasons contributing to the disparity in HTTPS adoption in China [12]. Restricted access to HTTPS websites such as Wikipedia and BBC can bring down the utilization of TLS-based services for China’s residents. Additionally, many of the most popular websites in China such as QQ and Baidu have a history of using HTTP over HTTPS. Since these sites demand a significant portion of the network traffic in China, they are large enough to impact the utilization statistics.

It can also be seen that China’s country-wide firewall is incompatible with the TLS style of encryption [13]. The firewall enables government regulation of illicit activity online such as violence and gambling, and it bolsters international cyber and network security. However, some of the filtering methods used require the ability to access the unencrypted content of network messages which could promote the under-utilization of TLS traffic.

5.2 Security Risks of Low HTTPS Adoption Rate

One way in which the results of this study could be used is to identify potential security risks and vulnerabilities to real-world attacks enabled by low levels of HTTPS utilization.

For example, the paper Security Analysis of the Chinese Web: How well is it protected? analyzes the implementation of insecure TLS/SSL protocols (among other web characteristics) [14]. Within this experiment, web-crawling was performed to acquire a representative sample of the Chinese web within a dataset of 10,000 websites. This study comparatively utilized a set of 6,400 candidate websites hosted within China. Security policies were analyzed and measured, and these results were compared to websites hosted outside of China to identify unique security risks.

5.3 Internet Security in China is Improving

There are also some notable developments in addressing the disparity of secure network algorithms being leveraged to protect data. In particular, the Cybersecurity Law of the People's Republic of China has been established as a step toward greater web security.

The article "New Chinese Cybersecurity and Data Privacy Requirements" from the Jones Day global law firm describes the impacts of the conditions within this new legislation [15]. Having taken effect in June 2017, this set of laws and regulations establishes national standards for network security in China. Encryption products in particular are receiving approval to aid in public interest, the national economy, and people's livelihoods as of January 2020 which is a helpful step towards greater web security.

5.4 Potential Next Steps and Future Work

This study provides an introductory look into the utilization of HTTPS in China, and future work can be pursued to continue expanding upon the findings made during this investigation. Some of

the areas worth looking into can include the following:

- Performing a similar type of analysis on countries with a comparable HTTPS history such as Japan or South Korea.
- Comparing HTTPS utilization differences at a finer scale such as comparing cities instead of comparing countries (for example, differentiating domains in Shenzhen vs Beijing) or exploring sub-domains or different paths under the same domain that may implement HTTP and HTTPS differently.
- Exploring related security topics such as utilization of tunnelling or cookies³.
- Comparing websites by industry or site content instead of geography⁴.
- Examining the TLS handshake itself using tools such as *scapy*⁵ instead of simply analyzing the end-result of the TLS connection [16].
- Exploring artifacts contained within the pages themselves such as links, images, and frames to content from other websites.

6 Conclusion

The research and analysis conducted in support of this study have contributed to the available collection of data and statistics associating HTTPS adoption with geographic regions, particularly in China. The findings of this study correlate with prior work in a conclusive determination that domains within China in 2023 still appear to be below-average in characteristics of HTTPS utilization and implementation compared to the rest of the world. Connections have been made between the results within this paper and similar resources which have performed familiar measurements, analyzed the distinct characteristics of network security which provide the most value, and considered the circumstances beyond the internet that influence such findings.

³ The exploration of HTTPS security within cookies was one of the focus areas within Security Analysis of the Chinese Web: How well is it protected? [14].

⁴ Comparing HTTPS adoption across industry was similarly explored as a means for comparison within Security Analysis of the Chinese Web: How well is it protected? [14].

⁵ This was partially explored within the experiments conducted in this study; however, it fell out-of-scope as it involved a different data collection method from the other tools leveraged. Hence, the results from that portion of the experiments are not presented here.

The network security landscape for domains located within China appears to be ever changing, and the trend toward a wider, faster, and safer internet continues moving forward.

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